

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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Gate Solution-2012

- Q.1 A 30 m steel tape having an area of cross-section of $5 \times 10^{-6} \text{ m}^2$ is standardized at 20°C , supported under a tension of 5.45 N. It is used to measure a horizontal distance of 81.15 m under an applied tension of 9.09 N. The error, due to incorrect pulling arrangement in this observation, in m is

$(E_{\text{steel}} = 200 \text{ GPa})$

- (A) 0.148 (B) 0.295 (C) 1.820 (D) 3.640

Answer (B)

Solution.

The error due to incorrect pulling arrangement is given by

$$C_p = \frac{(P - P_o)L}{AE}$$

Given that,

$$P = 9.09 \text{ N}; P_o = 5.45 \text{ N}; L = 81.15 \text{ m}$$

$$A = 5 \times 10^{-6} \text{ m}^2; E = 200 \times 10^9 \text{ N/m}^2$$

Now

$$C_p = \frac{(9.09 - 5.45) \times 81.15}{5 \times 10^{-6} \times 200 \times 10^9}$$

$$C_p = 2.95 \times 10^{-4} \text{ m}$$

$$C_p = 0.295 \text{ m.}$$

Q.2 The coefficient of variation of a dataset is measured by

- (A) $\frac{\text{mean}}{\text{standard deviation}}$ (B) $\frac{\text{mean}}{\text{variance}}$ (C) $\frac{\text{standard deviation}}{\text{mean}}$ (D) $\frac{\text{variance}}{\text{mean}}$

Answer (C)

Q.3 The value of $\int_0^1 \sin^{-1}(\cos x) dx$ is

- (A) $\frac{(\pi-1)}{2}$ (B) $\frac{(\pi+1)}{2}$ (C) $\frac{(2\pi+1)}{2}$ (D) $\frac{(2\pi-1)}{2}$

Answer (A)

Solution.

$$I = \int_0^1 \sin^{-1}(\cos x) dx$$

$$I = \int_0^1 \sin^{-1} \left\{ \sin \left(\frac{\pi}{2} - x \right) \right\} dx \quad \because \left[\sin \left(\frac{\pi}{2} - x \right) = \cos x \right]$$

$$I = \frac{\pi}{2} - \left[\frac{x^2}{2} \right]_0^1$$

$$I = \frac{\pi}{2} - \frac{1}{2} = \frac{\pi-1}{2}$$

Q.4 Assuming $\sin(1) = 0.841$ and $\sin(3) = 0.141$, the Lagrangian linear interpolating polynomial, for the function $f(x) = \sin(x)$ defined on the interval $[1, 3]$ and passing through the end points of the interval, is

(A) $-0.35x + 1.19$

(B) $-3.05x + 11.92$

(C) $-35.00x + 119.10$

(D) $-40.50x + 219.19$

Answer (A)

Solution .

Lagrangian Interpretation Formula

$$f(x) = \frac{(x-x_1) \dots (x-x_n)}{(x_0-x_1) \dots (x_0-x_n)} f_0 + \dots + \frac{(x-x_0) \dots (x-x_{n-1})}{(x_n-x_0) \dots (x_n-x_{n-1})} f_n$$

Given, $f(x) = \sin x$; $f_0 = 0.841$; $f_1 = 0.141$

$$x_0 = 1; x_1 = 3$$

$$\sin x = \frac{(x-3)}{(1-3)} \times 0.841 + \frac{(x-1)}{(1-3)} \times 0.141$$

$$\sin x = \frac{(x-3)}{-2} \times 0.841 + \frac{(x-1)}{2} \times 0.141$$

$$\sin x = -(x-3) \times 0.4205 + (x-1) \times 0.0705$$

$$\sin x = -0.4205x + 1.2615 + 0.0705x - 0.0705$$

$$\sin x = -0.3500x + 1.191$$

Q.5 If Poisson's ratio of a rock sample is 0.25, then the relationship among the modulus of Elasticity (E), modulus of rigidity (G) and bulk modulus (K) is

(A) $E = K = G$

(B) $E > G > K$

(C) $E = G > K$

(D) $E > K > G$

Answer (D)

Solution.

Given, Poisson ratio = $\frac{1}{m} = 0.25$

Or $m = 4$

Now, $K = \frac{mE}{3(m-2)}$

Where, K – Bulk modulus

E – Modulus of Elasticity

$$K = \frac{4E}{3(4-2)}$$

$$K = \frac{4E}{3 \times 2} = \frac{2}{3}E$$

Or $E = 1.5K$ (i)

Now $G = \frac{mE}{2(m+1)}$

where G – Modulus of rigidity

$$G = \frac{4E}{2(4+1)} = \frac{4E}{2 \times 5} = \frac{2}{5}E$$

$E = 2.5 G$ (ii)

From (i) & (ii)

$$1.5K = 2.5G$$

$$\therefore \frac{K}{G} = \frac{2.5}{1.5}$$

$\therefore K = 1.6 G$ (iii)

Hence, from equation (i), (ii) & (iii) it is seen that,

$$E > K > G$$

Q.6 The 2nd order differential equation having a solution $y = (A/x) + B$, where A and B are constants, is

(A) $\frac{d^2y}{dydx} + \frac{2dy}{x dx} = 0$

(B) $\frac{d^2y}{dx^2} + \frac{2dy}{x dx} = 0$

(C) $\left(\frac{d^2y}{dx^2}\right)^2 + \frac{2dy}{x dx} = 0$

(D) $\frac{d^2y}{dydx} + \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Answer (B)

Solution.

Given differential equation

$$y = \frac{A}{x} + B$$

Now differentiate w.r.t, x, we get

$$\frac{dy}{dx} = -\frac{A}{x^2} \dots\dots\dots(i)$$

Now differentiate equation (i). w.r.t , x, we get

$$\frac{d^2y}{dx^2} = \frac{2A}{x^3} \dots\dots\dots(ii)$$

Now multiplying (i) by $\frac{2}{x}$ we get,

$$\frac{2dy}{x dx} = -\frac{2A}{x^3} \dots\dots\dots(iii)$$

Now , adding (ii) & (iii) we get,

$$\frac{2dy}{x dx} + \frac{d^2y}{dx^2} = \frac{2A}{x^3} - \frac{2A}{x^3}$$

$$\frac{2}{x} \frac{dy}{dx} + \frac{d^2y}{dx^2} = 0$$

Q.7 A cylindrical rock specimen is uniaxially loaded under compression and fails at 50 MPa. The fracture plane is inclined at an angle of 45° with the axial direction. The normal and shear stresses respectively on the failure plane in MPa are

- (A) 50, 50 (B) 0, 50 (C) 50, 0 (D) 25, 25

Answer (D)

Solution.

Shear stress is given by-

$$\tau = \frac{\sigma_1}{2} = \frac{50}{2} = 25 \text{ MPa}$$

$$\text{Now, } \tau = \sigma_n \tan \phi + C$$

When $C = 0$;

$$\therefore \tau = \sigma_n \tan \phi$$

Given that, $\phi = 45^\circ$

$$\therefore 25 = \sigma_n \tan 45^\circ$$

$$\therefore \tau = \sigma_n \tan 45^\circ$$

$$\therefore \sigma_n = 25 \text{ MPa}$$

Q.8 In a surface mine, sound pressure level at a location generated by operation of a dozer and a drill respectively are 80 dBA and 60 dBA, when operated independently. The sound pressure generated by the dozer compared to the drill is higher by a factor of

- (A) 10 (B) 20 (C) 100 (D) 200

Answer (A)

Q.9 As per the Indian Electricity Rules 1956, the maximum permissible length of a flexible cable used with an electric rope shovel in m is

- (A) 100 (B) 200 (C) 300 (D) 500

Answer (C)

Q.10 The equipment that is NOT used in hard rock metal mining drivage is

- (A) road header (B) drill jumbo (C) jack hammer (D) dint header

Answer (D)

Q.11 The roof bolt that follows the principle of point anchorage is

- (A) expansion shell bolt (B) full column grouted bolt
(C) split set bolt (D) swellex bolt

Answer (A)

Q.12 Equipment used in mining of placer deposits is

- (A) auger (B) wagon drill (C) rope saw (D) riffle box

Answer (D)

Q.13 A dump truck powered by 350 kW engine is running at a speed of 35 km/h. Considering the transmission efficiency of the truck as 85%, the rim pull of the truck in kN is

- (A) 21 (B) 31 (C) 41 (D) 51

Answer (B)

Solution.

$$\text{Power} = \frac{\text{work done}}{\text{time}}$$

$$\text{Or power} = \frac{\text{rim pull} \times \text{distance}}{\text{time}}$$

$$\text{Or power} = \text{rim pull} \times \text{speed}$$

$$350 \times 10^3 = \text{Rim pull} \times \frac{35 \times 10^3}{60 \times 60}$$

$$\text{Rim pull} = \frac{350 \times 10^3 \times 60 \times 60}{35 \times 10^3}$$

$$\text{Rim pull} = 36000 \text{ N}$$

But given transmission efficiency is 85%

$$\text{Rim pull} = 36000 \times 0.85$$

$$\text{Rim pull} = 30600 \text{ N}$$

$$\text{Or Rim pull} = 30.6 \text{ KN or } 31 \text{ KN}$$

Q.14 Nystagmus is a miner's disease associated with

- (A) lever (B) lung (C) eye (D) stomach

Answer (C)

Q.15 Apart from mining of coal, the longwall mining method has been practiced for mining the deposits of

- (A) copper (B) lead and zinc
(C) manganese (D) pyrite and phosphate

Answer (D)

The three segments, whose synchronous functioning is essential for GPS operations, are

- (A) space, control and user (B) signal, control and user
(C) space, control and geo-registration (D) signal, control and geo-registration

Answer (A)

When a double ended ranging drum shearer cuts coal in a longwall face,

- (A) both the drums rotate in the same direction keeping the front drum up and the rear drum down
(B) both the drums rotate in the opposite direction keeping the front drum up and the rear drum down
(C) both the drums rotate in the opposite direction keeping the front drum down and the rear drum up
(D) both the drums rotate in the same direction keeping the front drum down and the rear drum up

Answer (B)

The match the following

Mine gas Principal constituent

P. Stink damp 1. CO

Q. White damp 2. H_2S

R. Black damp 3. CH_4

S. Fire damp 4. CO_2

(A) P-1, Q-2, R-3, S-4 (B) P-3, Q-4, R-1, S-2

(C) P-2, Q-1, R-4, S-3 (D) P-2, Q-1, R-3, S-4

Answer (C)

Q.19 Continuous miner and shuttle car combination is NOT applicable in mining with

- (A) rib pillar extraction technique
- (B) Wangawilli system
- (C) room and pillar method
- (D) longwall method

Answer (D)

Q.20 Contours in a topographic map

- (A) are not closed upon themselves although the earth is a continuous surface
- (B) are not perpendicular to the direction of maximum slope
- (C) provide an indication of presence of valley or ridge in the area
- (D) are the lines joining the points of same declination at different elevations

Answer (C)

Q.21 A Dragger Gas Mask DOES NOT filter

- (A) water vapour
- (B) nitrous fumes
- (C) carbon monoxide
- (D) carbon dioxide

Answer (B)

Q.22 A system consists of four elements A, B, C and D which are connected functionally in a parallel configuration. The individual reliability of the elements is 0.80, 0.82, 0.85 and 0.90 respectively. The reliability of the system is

- (A) 0.498
- (B) 0.602
- (C) 0.750
- (D) 0.999

Answer (D)

Solution.

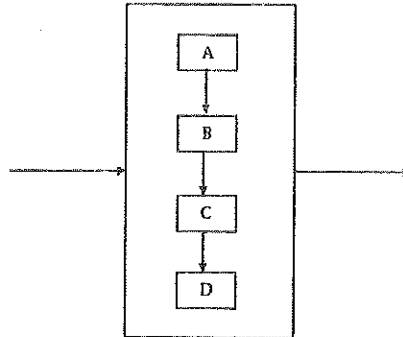
Given that,

Reliability of element A = 0.80

Reliability of element B = 0.82

Reliability of element C = 0.85

Reliability of element D = 0.90



The reliability of system will be

$$R = 1 - (1 - A) \cdot (1 - B) \cdot (1 - C) \cdot (1 - D)$$

$$R = 1 - (1 - 0.80) \cdot (1 - 0.82) \cdot (1 - 0.85) \cdot (1 - 0.90)$$

$$R = 0.999 \text{ Ans.}$$

Q.23 The blasting technique used for controlled throw of overburden is known as

- (A) cast blasting (B) coyote blasting
(C) plaster shooting (D) pop shooting

Answer (A)

Q.24 The stoping method, where a large part of blasted ore is allowed to accumulate in the stope to serve the purpose of providing working platform for stoping as well as to support the wall-rock, is known as

(A) shrinkage stoping

(B) cut and fill stoping

(C) square-set stoping

(D) sublevel stoping

Answer (A)

Q.25 The injury rates of mine workers in an underground coal mine based on age group are given below:

Age of mine workers	Age specific injury rate per 1000 persons	Age specific population in the mine
18-32	1.8	1000
33-46	2.5	500
47-60	4.5	300

The injury rate per 1000 persons employed in the mine for the total population is

(A) 0.24

(B) 2.44

(C) 8.80

(D) 24.40

Answer (B)

Solution.

Injury rate per 1000 employed in the mine for the total population is –

$$\begin{aligned} &= \frac{1.8 \times 1000 + 2.5 \times 500 + 4.5 \times 300}{1000 + 500 + 300} \\ &= \frac{4400}{1800} = 2.44 \end{aligned}$$

Q.26 A shearer is deployed in a mine where the specific energy consumption for cutting coal is 800 kJ/m^3 . The specific gravity of coal is 1.2. If the machine produces 700 te/h, the electrical power consumption in kW of the shearer at 65% motor efficiency is

(A) 149.4

(B) 199.4

(C) 219.4

(D) 239.4

Answer (B)

Solution.

$$\therefore 1 \text{ m}^3 \text{ coal consume} = 800 \text{ KJ}$$

$$1 \times 1.2 \text{ tonne of coal consume} = 800 \text{ KJ}$$

$$1 \text{ tonne of coal consume} = \frac{800}{1.2} \text{ KJ}$$

$$700 \text{ tonne of coal consume} = \frac{700 \times 800}{1.2} \text{ kJ}$$

$$700 \frac{\text{tonne}}{\text{h}} \text{ of coal consume} = \frac{700 \times 800}{1.2} \frac{\text{kJ}}{\text{h}}$$

$$\text{Or } 700 \frac{\text{tonne}}{\text{h}} \text{ of coal consume} = \frac{700 \times 800}{1.2 \times 3600} \frac{\text{kJ}}{\text{s}}$$

$$\text{Or } 700 \frac{\text{tonne}}{\text{h}} \text{ of coal consume} = \frac{700 \times 800}{1.2 \times 3600} \frac{\text{kJ}}{\text{s}}$$

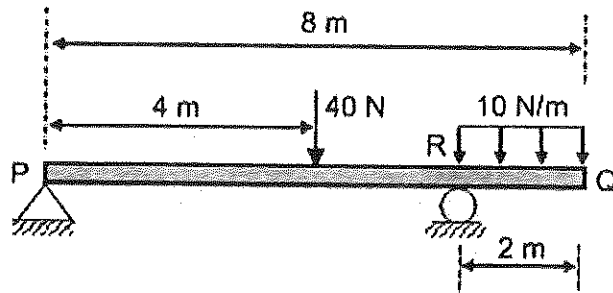
$$\text{Or } 700 \frac{\text{tonne}}{\text{h}} \text{ of coal consume} = 129.63 \text{ kW}$$

$$\therefore \text{Efficiency (\%)} = \frac{\text{output power}}{\text{input power}}$$

$$\therefore \text{Input power} = \frac{129.63}{0.65} = 199.4 \text{ kW}$$

Q.27

The figure shows a weightless beam PQ of length 8 m resting on a hinge support at P and on a roller support at R. A vertical force of 40 N is acting at a distance of 4 m from P. A uniformly distributed load of 10 N/m is acting on a length of 2 m of the beam from Q.



The magnitude of reaction force at R in N is

(A) 20

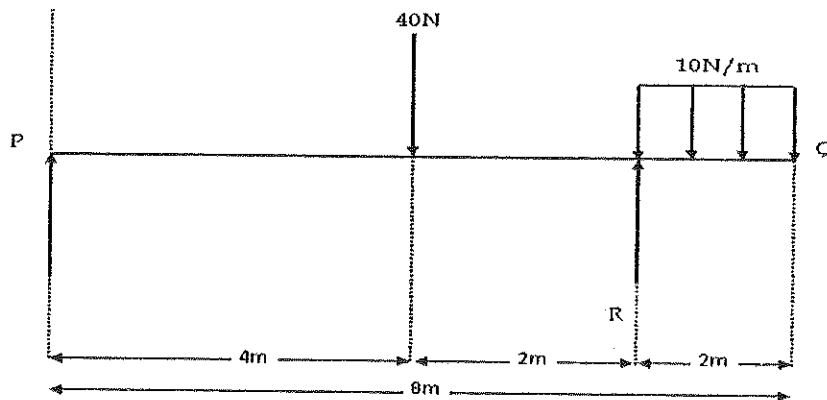
(B) 30

(C) 40

(D) 50

Answer (D)

Solution.



$$\sum M_P = 0$$

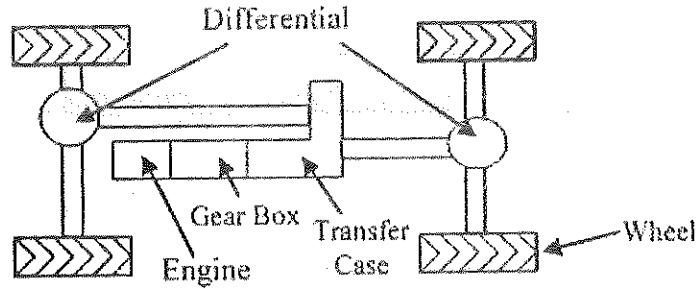
$$-40 \times 4 + R \times 6 - 10 \times 2 \times 7 = 0$$

$$\therefore R = \frac{140 + 160}{6}$$

$$\therefore R = 50 \text{ N}$$

Q.28 The gear ratios of the first gear, transfer case and differential of a four wheel drive vehicle are 3.81:1, 2.72:1 and 4.11:1 respectively. If the engine is rotating at 1000 rpm and the

wheel diameter is 1.2 m, the speed of the vehicle in first gear in km/h is



(A) 5.31

(B) 3.68

(C) 2.42

(D) 1.68

Answer (A)

Solution.

Circumstance of wheel = πD

$$= 3.14 \times 1.2 = 3.77 \text{ m}$$

Engine rpm = 1000 rpm

We have,

$$\text{Vehicle speed} = \frac{(\text{Engine rpm}) \times \text{wheel circumference}}{\text{combined gear ratio}}$$

$$\text{Vehicle speed} = \frac{1000 \times \pi \times 1.2 \times 1 \times 1 \times 1}{3.81 \times 2.72 \times 4.11}$$

$$= 88.47 \text{ m/minute}$$

$$= \frac{88.47 \times 60}{1000} \frac{\text{km}}{\text{h}}$$

$$= 5.31 \text{ Km/h}$$

Hence, speed of vehicle in first gear is -

$$= 5.31 \text{ Km/h}$$

Q.29

An iron ore mine recorded an average of 3 accidents per month. The number of

accidents is distributed according to Poisson distribution. The probability that there will be exactly 2 accidents per month is

- (A) 0.22 (B) 0.30 (C) 0.43 (D) 0.67

Answer (A)

Solution.

$$\text{Required probability} = \frac{e^{-m}(m)^r}{r!}$$

Given that, $m = 3$, $r = 2$

$$\text{Required probability} = \frac{e^{-3}(3)^2}{2!}$$

$$= 0.22$$

Q.30 Match the following:

Equipment	Component
P. Scraper	1. Dribble belt
Q. Dragline	2. Dipper stick
R. Bucket wheel excavator	3. Fair lead
S. Rope shovel	4. Bowl
(A) P-2, Q-4, R-3, S-1	(B) P-4, Q-2, R-1, S-3
(C) P-4, Q-3, R-1, S-2	(D) P-2, Q-4, R-1, S-3

Answer (C)

Q.31 The torque in N-m of a winder motor is described by the relationship $T = 1450 - 3.2\omega$, where, ω is the angular speed of the motor in rad/s. If the shaft is rotating at a speed of 1450 rpm, the power of the motor in kW is

(A) 112.4

(B) 146.4

(C) 184.4

(D) 212.4

Answer (B)

Solution.

$$\text{Given, } T = 1450 - 3.2 \omega$$

$$\omega = \frac{2\pi N}{60}, \text{ rad/sec}$$

Where, N - rpm

$$\omega = \frac{2 \times 3.14 \times 1450}{60} = 151.77 \text{ rad/second}$$

$$\text{Now, } T = 1450 - 3.2 \times 151.77$$

$$T = 964.336 \text{ Nm}$$

$$\text{Since } P = \omega T$$

$$P = 151.77 \times 964.336$$

$$P = 146.35 \text{ KW}$$

Q.32 An investment of Rs. 10,000, compounded annually, is estimated to return Rs. 20,000 after 6 years from the date of investment. The expected rate of return on this investment in percentage is

(A) 8.75

(B) 10.50

(C) 12.25

(D) 16.6

Answer (C)

Solution.

$$S = P(1+i)^n$$

$$\text{Given, } S = 20,000$$

$$P = 10,000$$

$$n = 6$$

$$\therefore 20,000 = 10,000 P (1+i)^n$$

$$2 = (1+i)^6$$

$$\therefore (2)^{1/6} = 1+i$$

$$\text{Or } i = (2)^{1/6} - 1$$

$$\therefore i = 12.25\%$$

Q.33 A spherical droplet of water, with density 1000 kg/m^3 and diameter of $1 \text{ }\mu\text{m}$, is falling in air. The viscosity of air is $1.85 \times 10^{-5} \text{ kg/m s}$. Neglecting air density and assuming that the settling of droplet in air follows Stokes' Law, the settling velocity in m/s is

- (A) 0.98×10^{-5} (B) 2.95×10^{-5} (C) 8.04×10^{-5} (D) 53.03×10^{-5}

Answer (B)

Solution.

According to Stoke's settling velocity is given by

$$v = \frac{g d^2 (\rho_p - \rho_f)}{18 \eta} \quad [\because \rho_f = 0]$$

$$v = \frac{9.81 \times (10^{-6})^2 \times 1000}{18 \times 1.85 \times 10^{-5}}$$

$$v = 2.95 \times 10^{-5} \text{ m/s}$$

Q.34 A mining company has three mines (M1, M2 and M3) that supply coal to three power plants (P1, P2 and P3). The three mines produce 900, 1000 and 1200 te of coal per day respectively. The power plant requirements from these three mines are 1200, 1000 and

900 te per day respectively. The unit cost of transporting coal from the three mines to the three power plants in Rs. is given below

		Power plants		
		P 1	P2	P3
Mines	M1	8	10	12
	M2	12	13	12
	M3	14	10	11

Based on the initial basic feasible solution, using Vogel's approximation method, the total transportation cost in Rs. is

- (A) 31200 (B) 31400 (C) 32800 (D) 40000

Answer (B)

Solution.

Given that,

		Power Plants			Supply
		P ₁	P ₂	P ₃	
Mines	M ₁	8	10	12	900
	M ₂	12	13	12	1000
	M ₃	14	10	11	1200
Requirement		1200	1000	900	Total = 3100

Now, for easy in calculation, taking scale of 100 = 1, for supply row and requirement column, then

8(9)	10	12	9/0(2)
12	13	12	10(1)
14	10	11	12(1)
12/3	10	9	31
(4)	(3)	(1)	
↑			

12	13	12	10(1)
14	10(10)	11	12/2(1)
3	10/0	9	22
(2)	(3)	(1)	
	↑		

12	12	10(0)
14	(2) 11	2/0(3) ←
3	9/7	12
(2)	(1)	

12(3)	7(12)	10/7/0
3/0	7/0	10

Therefore ,

8(900)	10	12	900
12(300)	13	12(700)	1000
14	10(1000)	11(200)	1200
1200	1000	900	3100

$$\begin{aligned}\text{Total transportation cost} &= 8 \times 900 + 12 \times 300 + 12 \times 700 + 10 \times 1000 + 11 \times 200 \\ &= 31400\end{aligned}$$

Hence total transportation cost is Rs. 31400

Q.35 The angle between the tangents to the curve $\vec{r} = t^2 \hat{i} + 2t \hat{j}$ at the point $t = \pm 1$ is

- (A) $\pi/2$ (B) $\pi/3$ (C) $\pi/4$ (D) $\pi/6$

Answer (A)

Solution.

$$\text{Given, } \vec{R} = t^2 \hat{i} + 2t \hat{j} \text{ and } t = \pm 1$$

$$\text{Now } \nabla \cdot \vec{R} = \left(\hat{i} \frac{\partial}{\partial t} + \hat{j} \frac{\partial}{\partial t} + \frac{\partial}{\partial t} \right) (t^2 \hat{i} + 2t \hat{j})$$

$$\nabla \cdot \vec{R} = 2t \hat{i} + 2 \hat{j}$$

Now at, $t = 1$

$$(\nabla \cdot \vec{R})_1 = 2\hat{i} + 2\hat{j}$$

Now at, $t = -1$

$$(\Delta.R)_2 = -2I + 2J$$

Since,

$$\cos\theta = \frac{(\nabla.R)_1 \cdot (\nabla.R)_2}{|\nabla.R|_1 \cdot |\nabla.R|_2}$$

$$\cos\theta = \frac{(2I+2J) \cdot (-2I+2J)}{\sqrt{2^2+2^2} \cdot \sqrt{(-2)^2+2^2}}$$

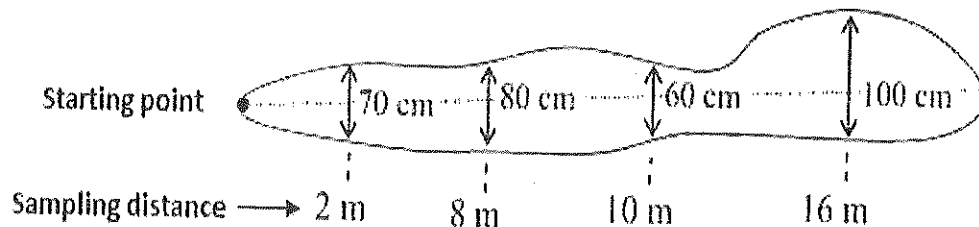
$$\cos\theta = \frac{-4+4}{\sqrt{8} \cdot \sqrt{8}}$$

$$\cos\theta = \frac{0}{8} = 0$$

$$\theta = \frac{\pi}{2}$$

- Q.36 The chip sampling data, spaced irregularly for a gold vein deposit, are shown in figure. The sample points have equal influence on both the sides.

Distance of sample from starting point (m)	2	8	10	16
Width (cm)	70	80	60	100
Assay (g/te)	5	7	6	4



The mean assay value in g/te is

- (A) 6.52 (B) 5.50 (C) 5.19 (D) 4.50

Answer (C)

Solution

Position of sample	Width (m)	Assay (g/te)	Distance of influence	Area of influence	Area assay product
2	0.7	5	5	3.5	17.5
8	0.8	7	4	3.2	22.4
10	0.6	6	4	2.4	14.4
16	1	4	6	6	24
		$\Sigma 22$	$\Sigma 19$	$\Sigma 15.1$	$\Sigma 78.3$

$$\begin{aligned}\text{Mean assay value} &= \frac{\Sigma \text{area assay product}}{\Sigma \text{area of influence}} \\ &= \frac{78.3}{15.1}\end{aligned}$$

Mean assay value = 5.19 g/tonnes

- Q.37 A series of triaxial tests of sandstone samples reveal the cohesion and the angle of internal friction as 21.65 MPa and 30° respectively. Based on the assumption that the sandstone samples follow the Mohr-Coulomb's failure criteria, the tensile strength in MPa is
- (A) 12.50 (B) 18.75 (C) 21.65 (D) 25.00

Answer (D)

Solution.

$$\begin{aligned}\text{Tensile strength } (\sigma_t) &= \frac{2c \cos \phi}{1 + \sin \phi} \\ \therefore \sigma_t &= \frac{2 \times 21.65 \times \cos 30^\circ}{1 + \sin 30^\circ} \\ \therefore \sigma_t &= 25 \text{ MPa}\end{aligned}$$

Q.38 The adjusted values of departure and latitude for a traverse line AB obtained in a field survey of a mine are 225.520 m and 388.835 m respectively. The length in m and azimuth of line AB are

- (A) 449.50, 30.11° (B) 614.36, 30.11°
(C) 614.36, 45.11° (D) 449.50, 45.11°

Answer (A)

Solution 40.

$$\text{Latitude} = l \times \cos\theta$$

$$\text{Departure} = l \times \sin\theta$$

Where, l = Length of line

θ = Azimuth of line

Given,

Latitude of line AB = 388.835 m

Departure of line AB = 225.520m

Now,

$$388.835 = l \times \cos\theta \dots\dots\dots(i)$$

$$225.520 = l \times \sin\theta \dots\dots\dots(ii)$$

Now dividing equation (ii) by equation (i),

$$\tan\theta = \frac{225.520}{388.835}$$

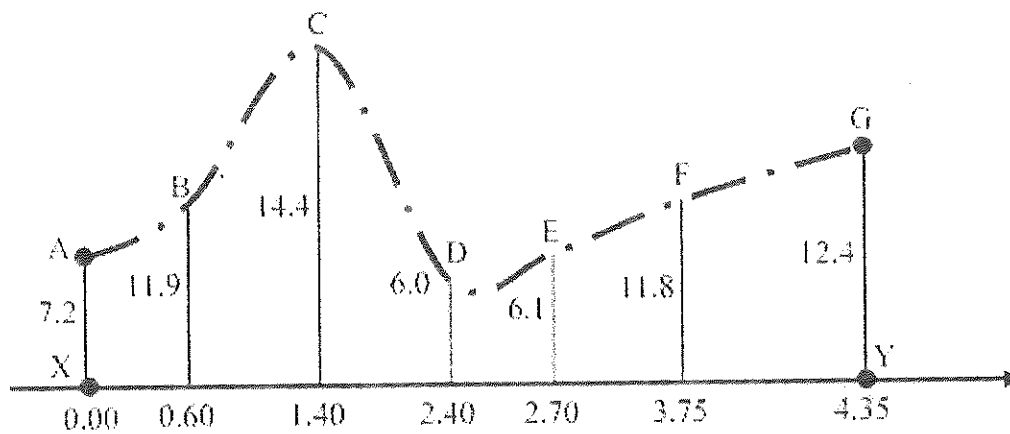
$$\theta = 30.11^\circ$$

Putting value of θ in equation (i), we get

$$388.835 = l \times \cos 30.11^\circ$$

$$\therefore l = 449.50 \text{ m}$$

- Q.39 The figure shows the values of seven perpendicular offsets and the respective locations along the line XY as observed while carrying out a traverse survey. The area of the plot XABCDEFGY in m^2 is



(All measurements are in meters)

- (A) 26.10 (B) 43.38 (C) 44.92 (D) 62.50

Answer (C).

Solution.

$$\text{The area of plot XABCDEFGY is} = \frac{1}{2} \times 0.6 \times (7.2 + 11.9) + \frac{1}{2} \times 0.8 \times (14.4 + 11.9) + \frac{1}{2} \times 1 \times$$

$$(14.4 + 6) + \frac{1}{2} \times 0.3 \times (6 + 6.1) + \frac{1}{2} \times 1.05 \times (11.8 + 6.1) + \frac{1}{2} \times 0.6 \times (11.8 + 12.4)$$

$$= 5.73 + 10.52 + 10.2 + 1.813 + 9.3975 + 7.26$$

$$= 44.92 \text{ m}^2$$

- Q 40. In a longwall panel, the main gate road is 1000 m long, 4.5 m wide and 2 m high. The gate road is to be used for airflow at the rate of $17 \text{ m}^3/\text{s}$. Considering a coefficient of resistance of airways of 0.01, the pressure in Pa required to maintain the airflow in the gate road is
- (A) 51.83 (B) 463.84 (C) 875.98 (D) 7885.32

Solution.

Pressure is given by

$$P = \frac{KSQ^2}{A^3}, \text{ Pa}$$

Given that,

$$K = 0.01, S = \text{Length} \times \text{Perimeter}$$

$$S = 1000 \times 2 \times (4.5 + 2)$$

$$S = 13000 \text{ m}^2$$

$$Q = 17 \text{ m}^3 / \text{s}$$

$$A = 4.5 \times 2 = 9 \text{ m}^2$$

$$\therefore P = \frac{0.01 \times 13000 \times (17)^2}{(9)^3}$$

$$\therefore P = 51.54 \text{ Pa}$$

- Q.41 The cofactor matrix of $P = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 3 & 1 \\ 1 & 2 & 3 \end{bmatrix}$ is

(A) $\begin{bmatrix} 21 & -5 & 2 \\ 2 & 21 & -5 \\ -5 & 2 & 21 \end{bmatrix}$ (B) $\begin{bmatrix} 21 & 2 & -5 \\ 2 & 7 & 15 \\ -5 & 21 & 2 \end{bmatrix}$

$$(C) \begin{bmatrix} -5 & 2 & 21 \\ 15 & 7 & 2 \\ -2 & 21 & -5 \end{bmatrix} \quad (D) \begin{bmatrix} 15 & 7 & 2 \\ -5 & 2 & 21 \\ -2 & 21 & -5 \end{bmatrix}$$

Solution.

$$\text{Cofactor of element 3 is } = (-1)^{1+1} \cdot \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = 7$$

$$\text{Cofactor of element 1 is } = (-1)^{1+2} \cdot \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = -5$$

$$\text{Cofactor of element 2 is } = (-1)^{1+3} \cdot \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1$$

$$\text{Cofactor of element 2 is } = (-1)^{2+1} \cdot \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = 1$$

$$\text{Cofactor of element 3 is } = (-1)^{2+2} \cdot \begin{vmatrix} 3 & 2 \\ 1 & 3 \end{vmatrix} = 7$$

$$\text{Cofactor of element 1 is } = (-1)^{2+3} \cdot \begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} = -5$$

$$\text{Cofactor of element 1 is } = (-1)^{3+1} \cdot \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} = -5$$

$$\text{Cofactor of element 2 is } = (-1)^{3+2} \cdot \begin{vmatrix} 3 & 2 \\ 2 & 1 \end{vmatrix} = 1$$

$$\text{Cofactor of element 3 is } = (-1)^{3+3} \cdot \begin{vmatrix} 3 & 1 \\ 2 & 3 \end{vmatrix} = 7$$

Hence, cofactor of Matrix P is $\begin{bmatrix} 7 & -5 & 1 \\ 1 & 7 & -5 \\ -5 & 1 & 7 \end{bmatrix}$

Q.42

Match the following:

Mining system

Face supports

P. Mechanized longwall in flat seam

1. Cable bolting

Q. Blasting gallery method

2. Shield support

R. Mechanized longwall in steep seam

3. Alpine breaker line support

S. Wangawilli method for 3 m thick coal seam

4. Troika shield support

(A) P-2, Q-1, R-4, S-3

(B) P-4, Q-1, R-3, S-2

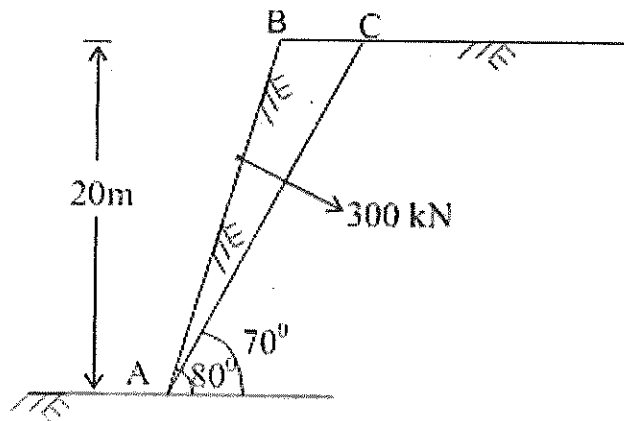
(C) P-4, Q-2, R-3, S-1

(D) P-2, Q-3, R-4, S-1

Answer (A)

Q.43

An opencast mine bench has a potential failure plane AC as indicated in figure. Bolts are installed to stabilize the failure plane providing a resultant bolting force of 300 kN. The area of sliding block ABC is 37.45 m². The unit weight, cohesion and angle of internal friction of rock are 25 kN/m³, 20 kPa and 40° respectively.



The factor of safety of slope when bolts are installed perpendicular to the failure plane is

(A) 0.79

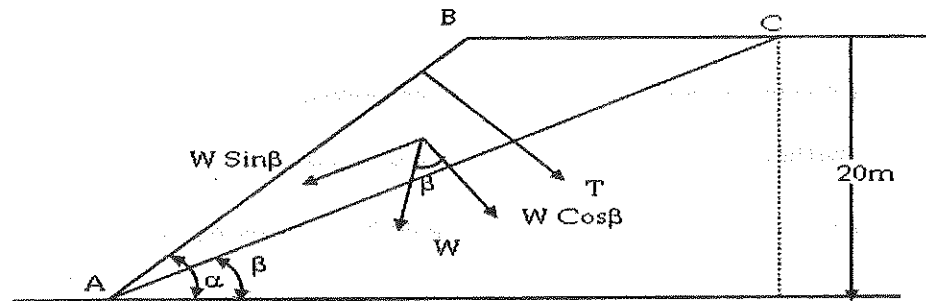
(B) 1.08

(C) 1.72

(D) 3.46

Answer (B)

Solution.



$$\therefore \text{FOS} = \frac{CA + (W \cos \beta + T) \tan \phi}{W \sin \beta}$$

where,

$$\text{Cohesion} = C = 20 \text{ KPa}; \alpha = 80^\circ; \beta = 70^\circ$$

$$\text{Shear area} = A = \left(\frac{20}{\sin \beta} \right) \times 1 = \left(\frac{20}{\sin 70^\circ} \right) \times 1, \text{ m}^2$$

$$W = 25 \times 37.45 \times 1 = 936.25 \text{ KN}$$

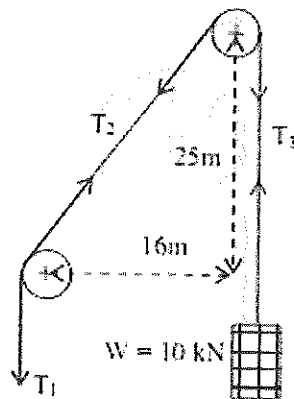
$$T = 300 \text{ KN} \text{ \& } \phi = 40^\circ$$

$$\therefore \text{FOS} = \frac{20 \times \left(\frac{20}{\sin 70^\circ} \right) + (936.25 \times \cos 70^\circ + 300) \tan 40^\circ}{936.25 \times \sin 70^\circ}$$

$$\therefore \text{FOS} = 1.08$$

Q.44 Figure shows a two pulley system for hoisting a load of 10 kN. The coefficient of friction between each pulley and the rope is 0.2. The vertical and horizontal distances between

the centers of the pulleys are 25 m and 16 m respectively.



The tensions T_1 and T_2 respectively in kN are

(A) 6.00, 5.38

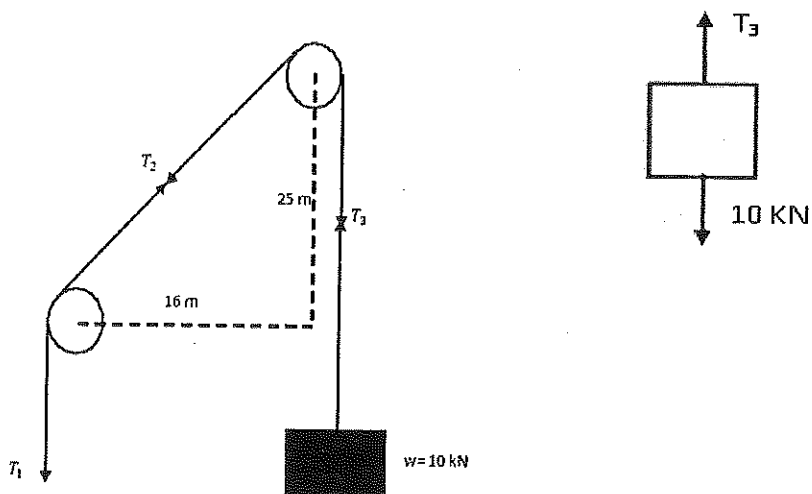
(B) 12.37, 11.06

(C) 18.74, 16.73

(D) 25.11, 22.41

Answer (C)

Solution.

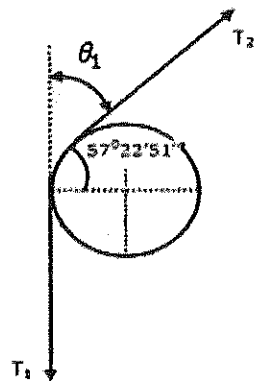


Here

Given , $\mu = 0.2$

$$\therefore T_3 - 10 = 0$$

$$T_3 = 10 \text{ kN}$$



$$\text{Now, } \tan \theta = \frac{25}{16}$$

$$\theta = 57^{\circ}22'51''$$

$\therefore$ Angle between T_1 & T_2 will be

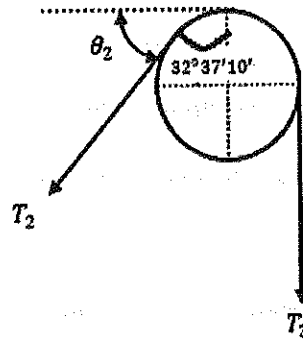
$$\theta_1 = 90^{\circ} - 57^{\circ}22'51'' = 32^{\circ}37'10''$$

$$\text{Or } \theta_1 = 0.57 \text{ radian}$$

$$\text{So, } \frac{T_1}{T_2} = e^{\mu \theta_1}$$

$$\frac{T_1}{T_2} = 2.718^{0.2 \times 0.57}$$

$$T_1 = 1.1207 T_2 \dots\dots\dots (i)$$



$$\tan \theta = \frac{16}{25}$$

$$\theta = 32^{\circ}37'10''$$

$\therefore$ Angle of lap between T_2 & T_3 is $= 90^{\circ} + \theta_2$

$$\left[\therefore \theta_2 = 90^{\circ} - 32^{\circ}37'10'' = 57^{\circ}22'51'' \right]$$

$\therefore$ Angle of lap between T_2 & T_3 is $= 90^{\circ} + 57^{\circ}22'51'' = 147^{\circ}22'51'' = 2.57$ radian

$$\text{So, } \frac{T_2}{T_3} = e^{\mu \times 2.57}$$

$$\frac{T_2}{10} = 2.718^{0.2 \times 2.57}$$

$$T_2 = 16.71 \text{ KN}$$

From equation (i),

$$T_1 = 1.1207 \times 16.71$$

$$T_1 = 18.73 \text{ KN}$$

Common Data for Questions 45 and 46:

A 2.5 m thick coal seam lying at an average depth of 100 m has been developed by bord and pillar method. The width of the square pillars is 30 m (centre to centre) and the gallery width is 4 m. The average density of the overlying strata is 26 kN/m³ and the pillar strength is 4500 kN/m².

Q.45 Extraction ratio during the development of the pillar is

(A) 0.129

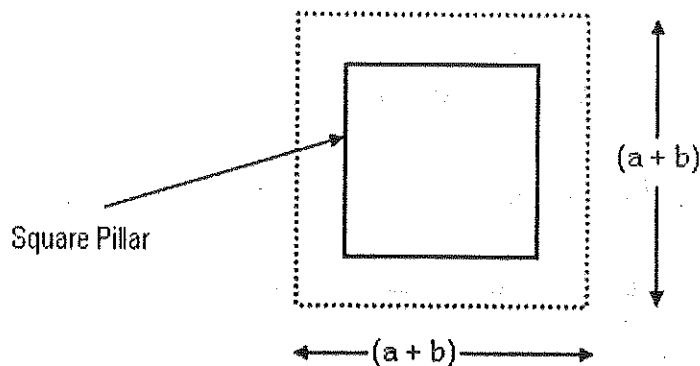
(B) 0.148

(C) 0.218

(D) 0.249

Answer (D)

Solution.



Given, $a + b = 30\text{m}$ and $b = \text{gallery width} = 4\text{m}$

$$a + b = 30$$

$$\therefore a = 30 - b = 30 - 4 = 26\text{m}$$

For square pillar extension ratio is given by-

$$R = 1 - \frac{a^2}{(a+b)^2}$$

$$R = 1 - \frac{(26)^2}{(26+4)^2} = 0.249$$

Q.46 The safety factor of the pillar is

(A) 1.1

(B) 1.3

(C) 1.5

(D) 1.7

Answer (B)

Solution.

$$\text{Factor of safety} = \frac{\text{strength of pillar}}{\text{Load acting on pillar}}$$

$$\text{Pillar strength is} = 4500 \text{ kN/m}^2 = 4500 \times 10^3 \text{ N/m}^2$$

Load acting on pillar is given by

$$P = P_0 \times \frac{1}{1-R}$$

$$\text{or } P = w.d \times \frac{1}{1-R}$$

$$w = 26 \text{ kN/m}^3 = 26 \times 10^3 \text{ N/m}^3$$

$$d = 100 \text{ m}$$

$$\text{Here, } P = 26 \times 10^3 \times 100 \times \frac{1}{1-0.249}$$

$$P = 3462050.599 \text{ N/m}^2$$

$$\text{So, } \text{FOS} = \frac{4500 \times 10^3}{3462050.599}$$

$$\text{FOS} = 1.3$$

Common Data for Questions 47 and 48:

The following data are provided for a surface mine to be excavated by a shovel:

Production target	: 10000 te/shift
Available hours per shift	: 6 hrs
Shovel loading cycles per hour	: 106
Bank density of the material mined	: 2400 kg/ m ³
Swing factor at 120 ° swing	: 0.91
Bucket fill factor	: 0.64
Utilization of available time	: 83%
No. of working days in a year	: 300
No. of shifts per day	: 3

Q.47 The annual production target in Mte is

- (A) 5.76 (B) 7.00 (C) 8.19 (D) 9.00

Answer (D)

Solution..

$$\begin{aligned}
 \text{Annual production target will be} &= 10000 \times 3 \times 300 \\
 &= 9000000 \text{ tonne} \\
 &= 9 \text{ Mte}
 \end{aligned}$$

Q.48 The size of bucket of the shovel in m³ is

- (A) 5.55 (B) 9.33 (C) 11.22 (D) 13.55

Answer (D)

Solution.

Productivity of Shovel : -

$$A = \frac{3600 \times Cd \times \alpha \times F \times \eta}{t_s}, m^3 / h$$

where,

Cd - Size of bucket, m^3

F - Fill factor

α - Swing factor

t_s - Cycle time, Sec

η - Percentage of utilization

Here,

$$\begin{aligned} A &= 10000 \text{ t/shift} = \frac{10000}{6} \text{ t/hour} \\ &= \frac{10000 \times 1000}{6} \text{ Kg/hour} \\ &= \frac{10000 \times 1000}{6 \times 2400} \text{ m}^3/\text{hour} \end{aligned}$$

$$\therefore t_s \rightarrow 1 \text{ hour} = 106 \text{ cycle}$$

$$106 \text{ cycle} = 1 \text{ hour}$$

$$1 \text{ cycle} = \frac{1}{106} \text{ hour} = \frac{3600}{106} \text{ seconds}$$

Putting all values,

$$\frac{10000 \times 1000}{6 \times 2400} = \frac{3600 \times Cd \times 0.64 \times 0.91 \times 1 \times 0.83}{\frac{3600}{106}}$$

$$Cd = \frac{10000 \times 1000}{6 \times 2400 \times 0.64 \times 0.91 \times 0.83 \times 10^6}$$

$$Cd = 13.55 \text{ m}^3$$

Statement for Linked Answer Questions 49 and 50:

A mining project is composed of five activities whose three time estimates in months are given below:

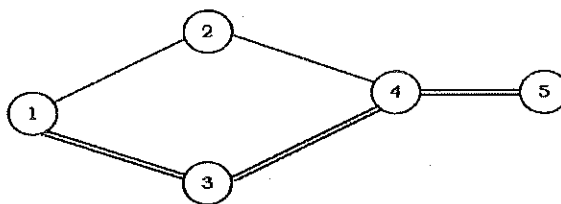
Activity	Estimated duration (months)		
	Optimistic time	Most likely time	Permissible time
1-2	1	1	7
1-3	2	5	8
2-4	1	1	7
3-4	2	5	14
4-5	3	6	15

Q.49 The expected duration of the mining project in months is

- (A) 5 (B) 16 (C) 18 (D) 29

Answer (C)

Solution.



Activity	t_0	t_m	t_p	$t_c = \frac{t_0 + 4t_m + t_p}{6}$	$\sigma^2 = \left[\frac{t_p - t_0}{6} \right]^2$
1-2	1	1	7	2	1
1-3	2	5	8	5	1
2-4	1	1	7	2	1
3-4	2	5	14	6	4
4-5	3	6	15	7	4

Hence various path through the network and their length is given by-

$$1-2-4-5 = 2 + 2 + 7 = 11$$

$$1-3-4-5 = 5 + 6 + 7 = 18$$

Since path 1-3-4-5 has longest duration, it is the critical path of network.

$\therefore$ Expected project length is = 18 months.

Q.50 The standard deviation of the project length in months is

(A) 2

(B) 3

(C) 6

(D) 9

Answer (B)

Solution.

$$\text{Variance} = V_{1-3} + V_{3-4} + V_{4-5}$$

$$V = 1 + 4 + 4 = 9$$

$$\text{But } V = \sigma^2$$

$$\sigma^2 = 9$$

$$\sigma = 3$$

Statement for Linked Answer Questions 51 and 52:

In a mine between upcast shaft and downcast shaft, two airways are connected in parallel and their resistances are $100 \text{ N s}^{-2} \text{ m}^{-8}$ and $120 \text{ N s}^{-2} \text{ m}^{-8}$ respectively. The resistance of upcast shaft, downcast shaft and the fan drifts are 10, 20 and $5 \text{ N s}^{-2} \text{ m}^{-8}$ respectively. The fan drift air pressure is 15 MN/m^2 .

Q.51 The rate of airflow through the mine in m^3/s is

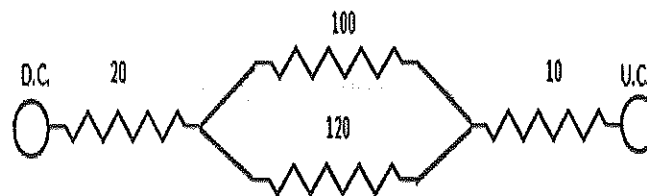
(A) 4.16

(B) 18.26

(C) 240.35

(D) 333.33

Solution.



$$\frac{1}{\sqrt{R}} = \frac{1}{\sqrt{R_1}} + \frac{1}{\sqrt{R_2}}$$

$$\frac{1}{\sqrt{R}} = \frac{1}{\sqrt{100}} + \frac{1}{\sqrt{120}}$$

$$R = 27.33 \text{ N s}^{-2} \text{ m}^{-8}$$

$$\text{Total resistance of mine} = 27.33 + 20 + 10 + 5$$

$$= 62.33 \text{ N s}^{-2} \text{ m}^{-8}$$

$$\text{Now } P = RQ^2$$

$$\text{Given, } P = 15 \text{ MN/m}^2$$

$$\therefore 15 \times 10^6 = 62.33 \times Q^2$$

$$Q = \sqrt{\frac{15 \times 10^6}{62.33}}$$

$$Q = 490.57 \text{ m}^3/\text{Sec}$$

- Q.52 The rate of airflow through the split airway having resistance of $100 \text{ N s}^{-2} \text{ m}^{-8}$ in m^3/s is
- (A) 0.42 (B) 0.79 (C) 2.19 (D) 7.90

Solution.

Rate of air flow through the split air way having resistance of $100 \text{ N s}^{-2} \text{ m}^{-8}$ will be

$$Q_1 = Q \times \sqrt{\frac{R}{R_1}}$$

$$Q_1 = 490.57 \times \sqrt{\frac{27.33}{100}}$$

$$Q_1 = 256.46 \text{ m}^3/\text{s}$$

- Q.53 Choose the most appropriate alternative from the options given below to complete the following sentence:

I ____ to have bought a diamond ring.

- (A) have a liking (B) should have liked
- (C) would like (D) may like

Answer (C)

- Q.54 Choose the most appropriate alternative from the options given below to complete the

Food prices ___ again this month.

- Answer (C)**

Answer (D)

Answer (A)

Answer (C)

Solution.

Let five natural numbers are x_1, x_2, x_3, x_4, x_5 , then

$$12 = \frac{x_1 + x_2 + x_3 + x_4 + x_5}{5}$$

$$\therefore x_1 + x_2 + x_3 + x_4 + x_5 = 60$$

Take, $x_1 = 50, x_2 = 4, x_3 = 3, x_4 = 2, x_5 = 1$

Then, $50 + 4 + 3 + 2 + 1 = 60$

$$60 = 60$$

Hence, largest possible value among the numbers is-

$$x_1 = 50$$

Q.58 Two policemen, A and B, fire once each at the same time at an escaping convict. The Probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

(A) 0.14

(B) 0.22

(C) 0.33

(D) 0.40

Answer (A)

Solution.

$$P(A) = 3 P(B) \quad \text{and} \quad P(A \cup B) = 0.5$$

$$\text{Since } P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{Here, } P(A \cap B) = 0$$

$$\therefore P(A \cup B) = P(A) + P(B)$$

$$0.5 = 3 P(B) + P(B)$$

$$4P(B) = 0.5$$

$$P(B) = \frac{0.5}{4}$$

$$P(B) = 0.14$$

Q.59 The total runs scored by four cricketers P, Q, R, and S in years 2009 and 2010 are given in the following table:

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

(A) P

(B) Q

(C) R

(D) S

Answer (B)

Solution.

For player P percentage increases in total runs

$$= \frac{1008 - 802}{802} \times 100 = 20.43\%$$

For player Q percentage increases in total runs

$$= \frac{912 - 765}{765} \times 100 = 16.12\%$$

For player R percentage increases in total runs

$$= \frac{619 - 429}{619} \times 100 = 30.70\%$$

For player S percentage increases in total runs

$$= \frac{701 - 501}{701} \times 100$$

$$= 28.53\%$$

Hence, the player with the lowest percentage is total runs is Q. Ans.

Q.60 If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

- (A) sum of squares of two natural numbers
- (B) sum of cubes of two natural numbers
- (C) sum of square roots of two natural numbers
- (D) sum of cube roots of two natural numbers

Answer (A)

Solution.

Let prime number is 5 and when this prime number is divided by 4 then gives remainder of 1.

Let two natural numbers are 2, 1 and hence prime number 5 can be expressed as sum of

squares of these two natural numbers i.e

$$2^2 + 1^2 = 5$$

$$4 + 1 = 5$$

$$5 = 5$$

Q.61 Two points (4, p) and (0, q) lie on a straight line having a slope of $\frac{3}{4}$. The value of (p – q) is

(A) -3

(B) 0

(C) 3

(D) 4

Answer (C)

Solution.

$$y = mx + C \dots\dots\dots(i)$$

Given two points (4, P) and (0, q) and slope of $\frac{3}{4}$

Now, at point (4, P) and slope $\frac{3}{4}$, equation (i) becomes

$$P = \frac{3}{4} \times 4 + C$$

$$P = 3 + C \dots\dots\dots(ii)$$

Now at point (0,q) and slope $\frac{3}{4}$, equation (i) becomes,

$$q = \frac{3}{4} \times 0 + C$$

$$q = C \dots\dots\dots(iii)$$

from equation (ii) and (iii) , it is found that

$$p - q = 3 + C - C$$

$$P - q = 3$$

Q.62 In the early nineteenth century, theories of social evolution were inspired less by Biology than by the conviction of social scientists that there was a growing improvement in social institutions. Progress was taken for granted and social scientists attempted to discover its laws and phases.

Which one of the following inferences may be drawn with the greatest accuracy from the

above passage?

Social scientists

(A) did not question that progress was a fact.

(B) did not approve of Biology.

(C) framed the laws of progress.

(D) emphasized Biology over Social Sciences.

Answer (A)